Week 3: Shuffle machines

Adult guide built from the problem files · Unpiloted.

The answers, hints and routes are in the adult guide, week-03-facilitator.pdf, whose source is guide-src/facilitator.tex. First route = the guide's core problems.

Problems and routes

Problem	K–1	2–3	4–5	route
Run each machine until every cube is home	P1	P1	P1	first
Predict the turns without moving cubes		P2	P2	first
Turn counts with 5 slots		P3	P3	first
Four cubes on two machines	P2			first
A 4-slot and a 5-slot machine	P3			first
Arrows for 2, 3 and 4 turns	P4			first
Only one cube out of place	P5			return
Draw the machine that sends every cube home	P6			return
All the machines with 3 slots	P7			return
A 5-slot machine that takes 6 turns	P8			return
Draw the machine that undoes one turn		P4		first
Machines that undo themselves		P5		return
A then B and B then A		P6		return
Turn counts with 6 slots		P7		return
More than 10 turns with 7 slots		P8		return
Slowest machines with 6 and 7 slots			P4	first
Perfect shuffles of 4 to 12 cards			P5	first
A then B and B then A for three pairs			P6	return
Undoing A then B			P7	return
Two machines with A then B equal to B then A			P8	return
Slowest machines for 1 to 10 slots			P9	return
Perfect shuffles of a full deck			P10	return

First route. K–1 P1–P4; Grades 2–3 P1–P4; Grades 4–5 P1–P5. **Return visits.** K–1 P5–P8; Grades 2–3 P5–P8; Grades 4–5 P6–P10.

Key, one entry per problem

These problem files hold no keys; the week's adult guide has them.